

DR. LIAM MOROY

Generative-AI researcher working on robust, uncertainty-aware solutions for scientific problems.

SUMMARY

Research Associate at Heriot-Watt University, developing deep generative models (diffusion / score-based) for severely ill-posed inverse problems, with a focus on robust, uncertainty-aware reconstruction. Experienced across the full research cycle — from problem formulation and algorithm development to large-scale multi-GPU training, publications, and open-source code.

EDUCATION

Ph.D. -- Université Paris-Saclay & ONERA The french aerospace lab

- Generative AI (diffusion models) for solving inverse problems in imaging, applied to tomographic imaging.

📅 2022 - 2025

📍 Paris, France

MSc Electrical Engineering + MSc Complex Systems

- Enseirb-Matmeca engineering school, Bordeaux INP and University of Bordeaux (concurrent).

📅 2020 - 2022

📍 Bordeaux, France

BSc - Electronics Engineering

- Enseirb-Matmeca engineering school, Bordeaux INP.

📅 2017 - 2020

📍 Bordeaux, France

RESEARCH ACTIVITIES

Selected Publications:

- “On the Posterior Gap in Plug&Play Diffusion Methods for Sparse-View CT”, IEEE JSTSP 2026 Special Issue on High-Dimensional Imaging: Emerging Challenges and Advances in Reconstruction and Restoration, [L. Moroy](#), G. Bourmaud, F. Champagnat, J.-F. Giovannelli.
- “Evaluating the Posterior Sampling Ability of Plug&Play Diffusion Methods in Sparse-View CT”, ICASSP 2025, [L. Moroy](#), G. Bourmaud, F. Champagnat, J.-F. Giovannelli.

Selected Oral Presentations:

- “Improving the Posterior Sampling of PnP Diffusion Models by Super-Calibration: application to Poisson inverse problem” 7th SLSIP Workshop, May 2026, Krujë, Albania.
- “Evaluating the Posterior Sampling Ability of Plug&Play Diffusion Methods in Sparse-View CT”, GdR IASIS: Advances in Learning-Based Image Restoration, Dec 2024, Henri Poincaré Institute, Paris.

TEACHING & COMMUNICATION

University of Bordeaux

- Master ISC – MSc teaching assistant - Introduction to noise and random signals (2023/24 & 2024/25).

ENSEIRB-MATMECA engineering school

- Engineering cycle – MSc teaching assistant - Generative modelling with diffusion models (2024/25).
- Engineering cycle – Freshman teaching assistant - Introduction to programming in C/C++ (2023/24 & 2024/25).

Interactive scientific outreach

- Interactive in-browser demos of generative modelling methods (annealed Langevin sampling, diffusion posterior sampling for inpainting), aimed at technical and non-specialist audiences.

PROFESSIONAL EXPERIENCE

Research Associate -- Generative AI for Inverse Problems in Physical Sciences

Heriot-Watt University (BISC)

📅 Dec. 2025 - Present

📍 Edinburgh, Scotland

- Development and application of deep generative models for solving highly ill-posed inverse problems across multiple physical imaging modalities: Computed Tomography for nuclear applications, non-conventional optics, and medical imaging.
- Collaboration within a multi-disciplinary team of statisticians, physicists, and engineers on high-impact scientific and medical imaging applications.

R&D Internship -- *Deep Learning* for tomographic reconstruction of compressible flows

ONERA (DTIS/MIC)

📅 Feb. 2022 - Aug. 2022

📍 Paris, France

- Design and evaluation of learned prior models for Bayesian CT reconstruction of compressible flows from limited measurements.

R&D Internship -- *Deep Learning* for nebula removal in astronomical images

Astrophysics Laboratory of Bordeaux (LAB) & IMS Laboratory

📅 Jul. 2021 - Sep. 2021

📍 Bordeaux, France

- Training a regressor from noisy ground truths and transfer learning from synthetic datasets.

SELECTED PROJECTS

Generative Modelling of Implied Volatility Surfaces

Independent project -- github.com/LiamMry/iv-surface-diffusion

📅 2026

- Score-based diffusion model (VP-SDE, attention U-Net) trained on $\sim 3,500$ daily SPY implied-volatility surfaces (2010--2023); no-arbitrage consistency emerges from data without explicit constraints.
- Frames daily surface calibration as posterior sampling under a diffusion prior; benchmarked ODE/SDE samplers (Euler, Heun, DPM-Solver).

SKILLS

- **Programming:** Python, C, C++
- **Machine Learning & Scientific Computing:** CUDA, GPU-Accelerated Computing, PyTorch, Slurm, HPC Clusters
- **Generative Modeling:** Score-Based Generative Modeling, Flow-Matching, Rectified Flow-Matching, Autoregressive modeling, Normalizing Flows, Variational Auto-Encoders
- **Research Methods & Domains:** Generative Models, Plug-and-Play Methods, Bayesian Inference, Inverse Problems, Uncertainty Quantification, Statistical Data Analysis, Experimental Design, Signal Processing, Image Processing
- **Computing & Research Tools:** Git, GitHub, Linux, Bash, LaTeX, Slurm, High-Performance Computing (HPC) Clusters
- **Languages:** French (Native), English (Fluent; TOEIC 950), Spanish (B1; Intermediate), Japanese (Beginner)